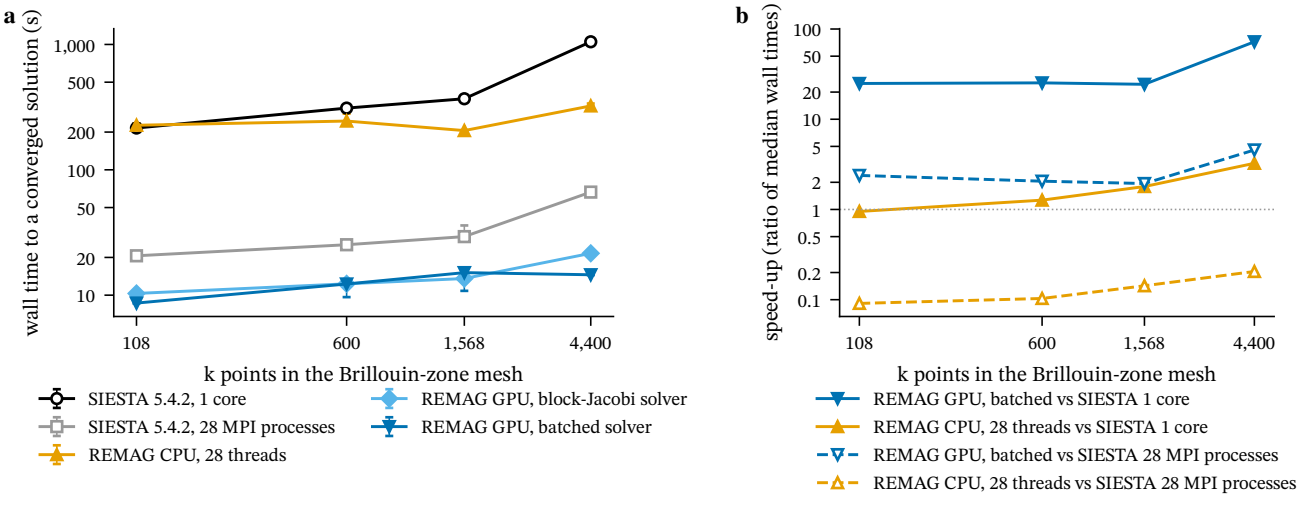
a
wall time to a converged solution (s)
1,000
500
200
100
50
20
10
108
600
1,568
4,400
k points in the Brillouin-zone mesh
SIESTA 5.4.2, 1 core
SIESTA 5.4.2, 28 MPI processes
REMAG CPU, 28 threads
REMAG GPU, block-Jacobi solver
REMAG GPU, batched solver
b
speed-up (ratio of median wall times)
100
50
20
10
5
2
1
0.5
0.2
0.1
108
600
1,568
4,400
k points in the Brillouin-zone mesh
REMAG GPU, batched vs SIESTA 1 core
REMAG CPU, 28 threads vs SIESTA 1 core
REMAG GPU, batched vs SIESTA 28 MPI processes
REMAG CPU, 28 threads vs SIESTA 28 MPI processes